

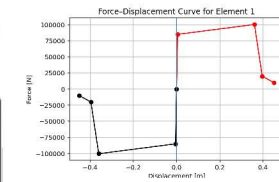
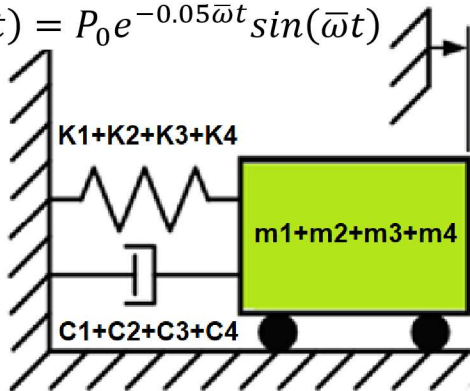
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

COMPREHENSIVE EVALUATION OF THE EQUIVALENT VISCOUS DAMPING RATIO OF A NONLINEAR SDOF SYSTEM WITH FOUR ELEMENTS STRUCTURE UNDER MULTIPLE LOADING PROTOCOLS

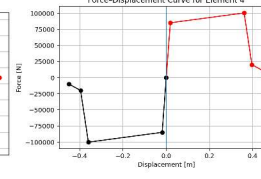
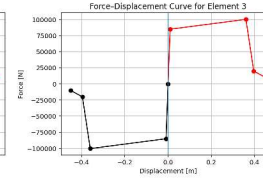
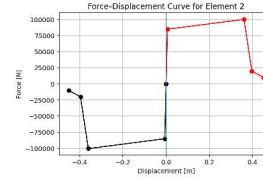
THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAIE (QASHQAI)

INCREMENTAL
DISPLACEMENT

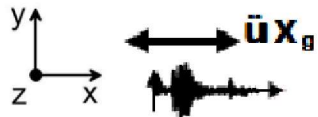
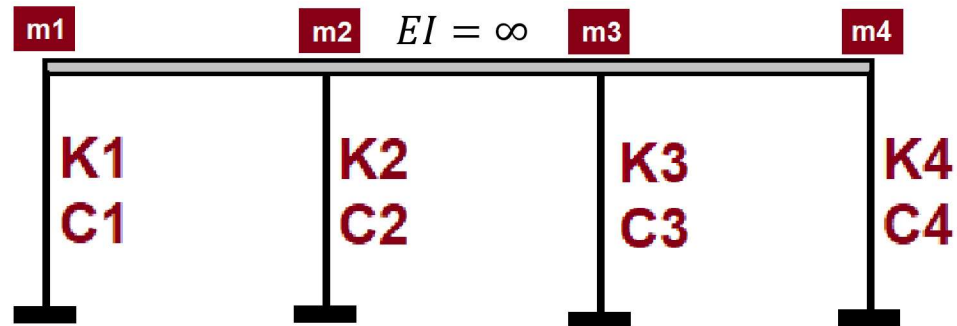
$$P(t) = P_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$



Initial Displacement
Initial Velocity
Initial Acceleration



INCREMENTAL
DISPLACEMENT



Energy quantities and the damping formula

- **Dissipated energy** E_d – the area enclosed by the hysteresis loop. This is the net work done per cycle and represents the energy that must be removed from the system to return it to the original state. Physically, it corresponds to plastic yielding, friction, cracking, or other irreversible mechanisms.
- **Maximum strain energy** E_s – defined here as

$$E_s = \frac{1}{2} F_{\max} u_{\max}$$

where $u_{\max} = \max |u|$ is the peak absolute displacement in the loop and $F_{\max}$ is the absolute value of the force **at that same displacement**. This definition assumes a **secant stiffness** $k_{\text{sec}} = F_{\max}/u_{\max}$ and represents the elastic strain energy stored at peak displacement if the system were linear with that stiffness.

The equivalent viscous damping ratio is then:

$$\xi_{eq} = \frac{100 E_d}{4\pi E_s} \quad [\%]$$

This stems from equating E_d to the energy dissipated per cycle by a linear viscous damper, $E_d = \pi c \omega u_{\max}^2$, while $E_s = \frac{1}{2} k u_{\max}^2$; substituting $\xi = c/(2m\omega)$ and $\omega^2 = k/m$ yields the familiar expression $\xi = E_d/(4\pi E_s)$.

$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

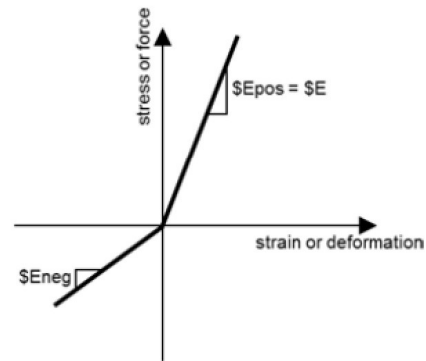
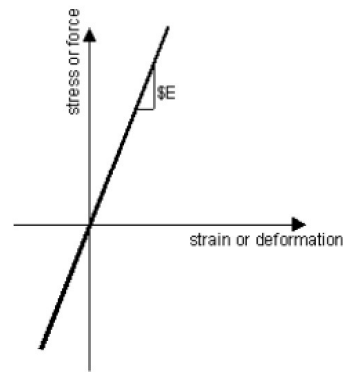
Δ_d = Lateral Displaement from Dynamic Analysis

Δ_y = Lateral Yield Displaement from Pushover Analysis

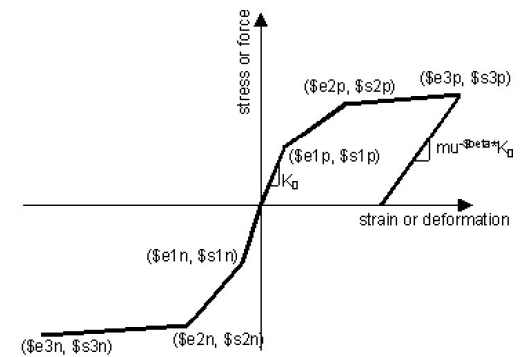
Δ_u = Lateral Ultimate Displaement from Pushover Analysis

$$P(t) = P_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$

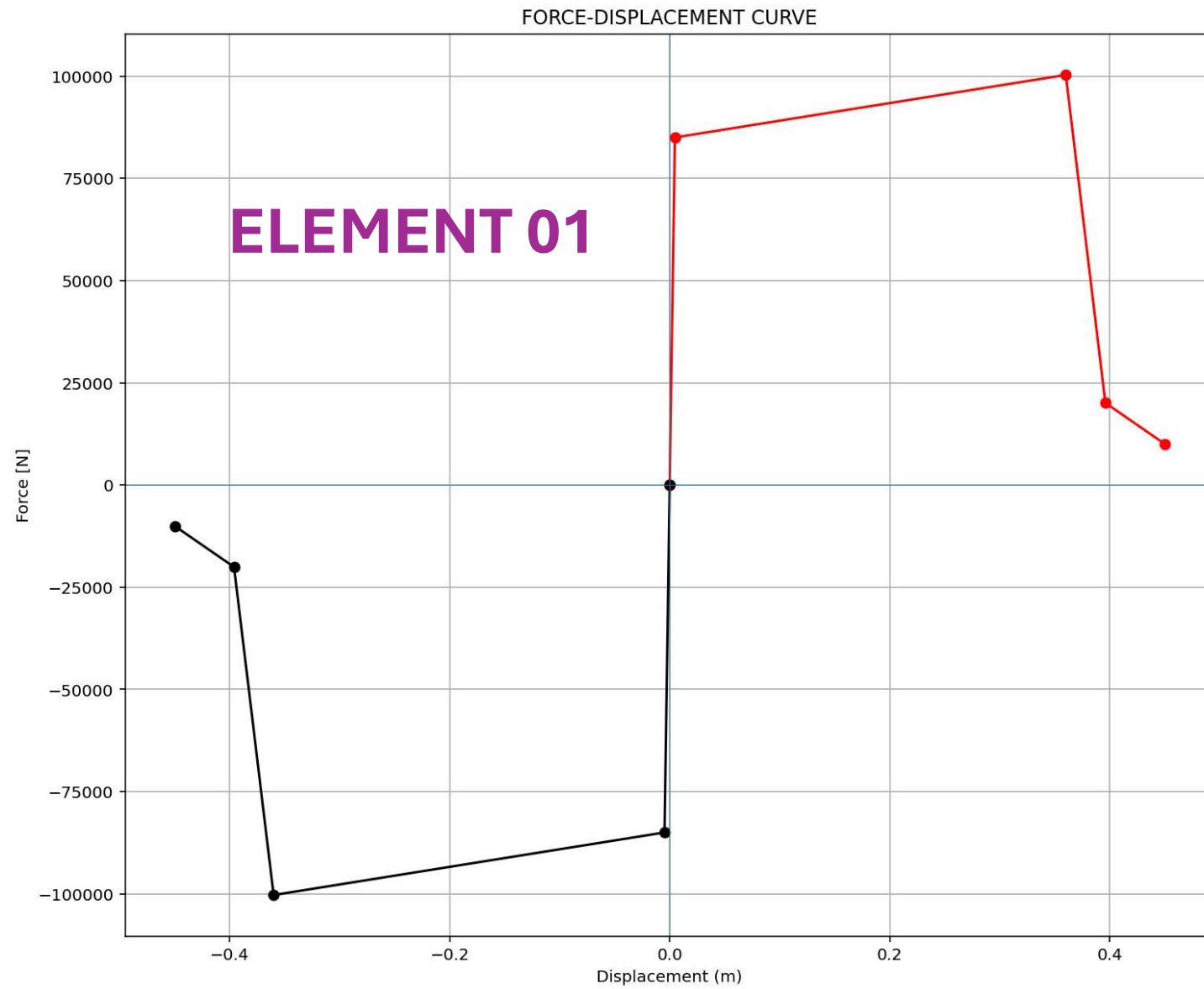
$$m\ddot{u} + c\dot{u} + ku = P(t)$$

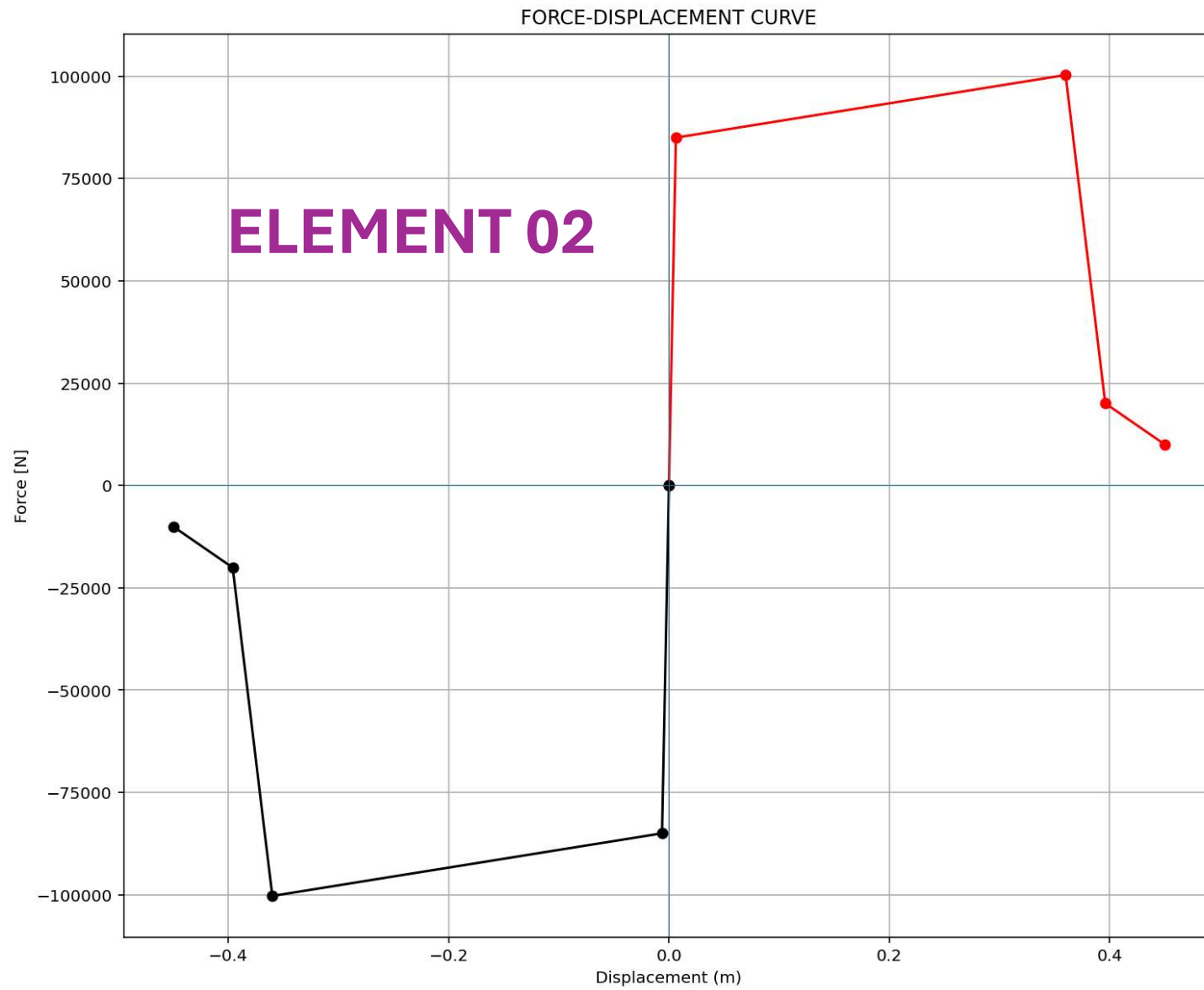


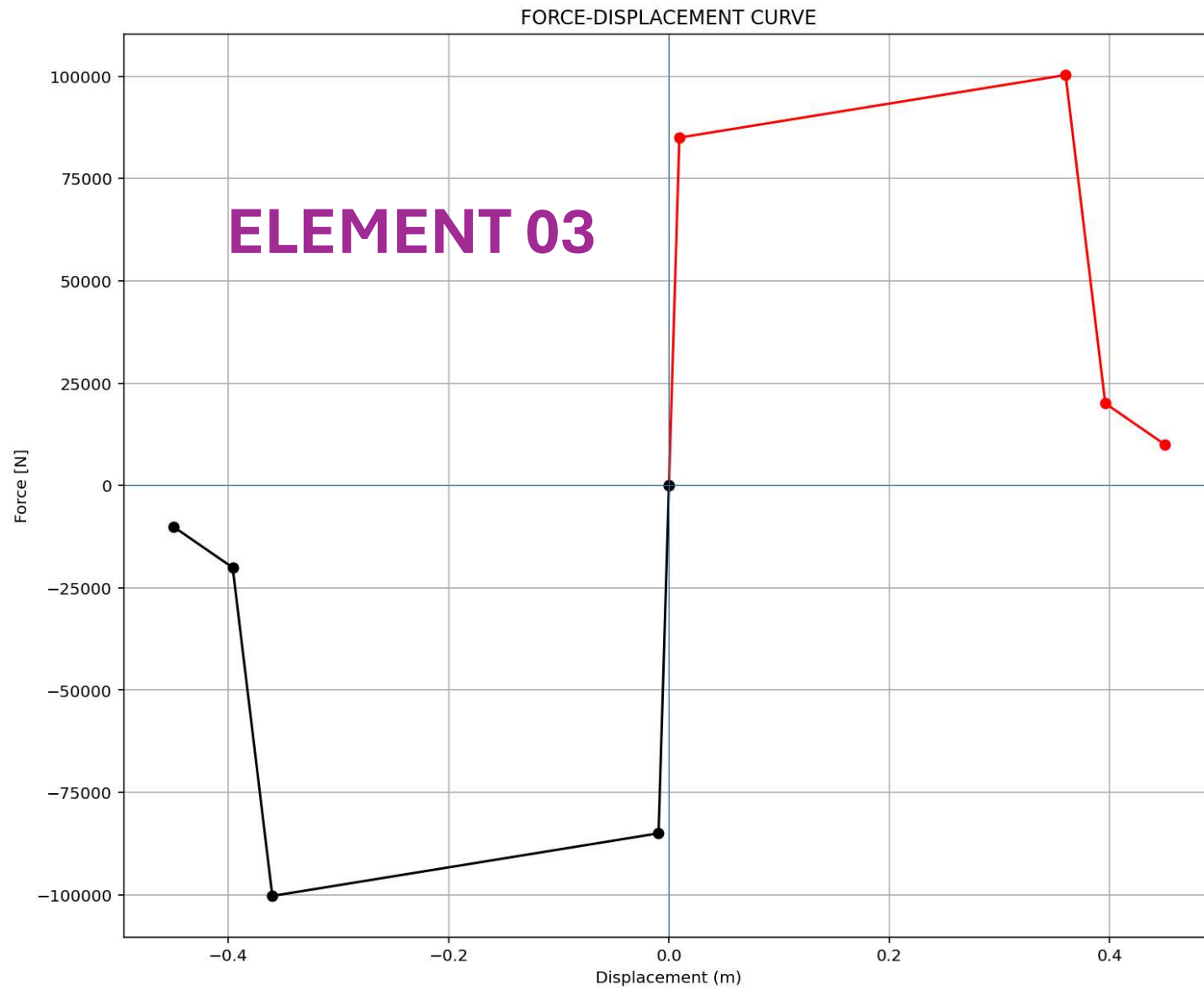
ELASTIC MATERIAL

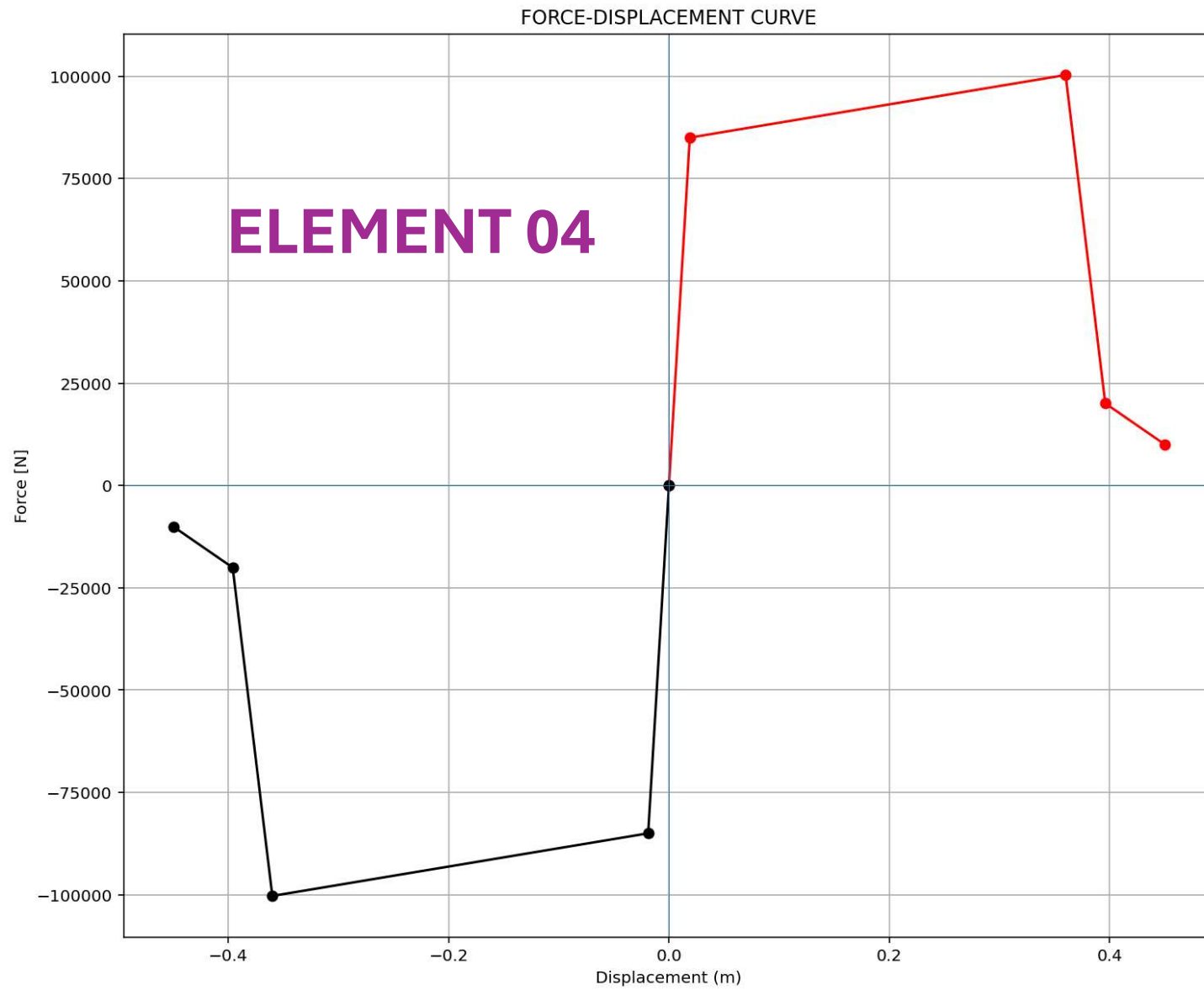


INELASTIC MATERIAL



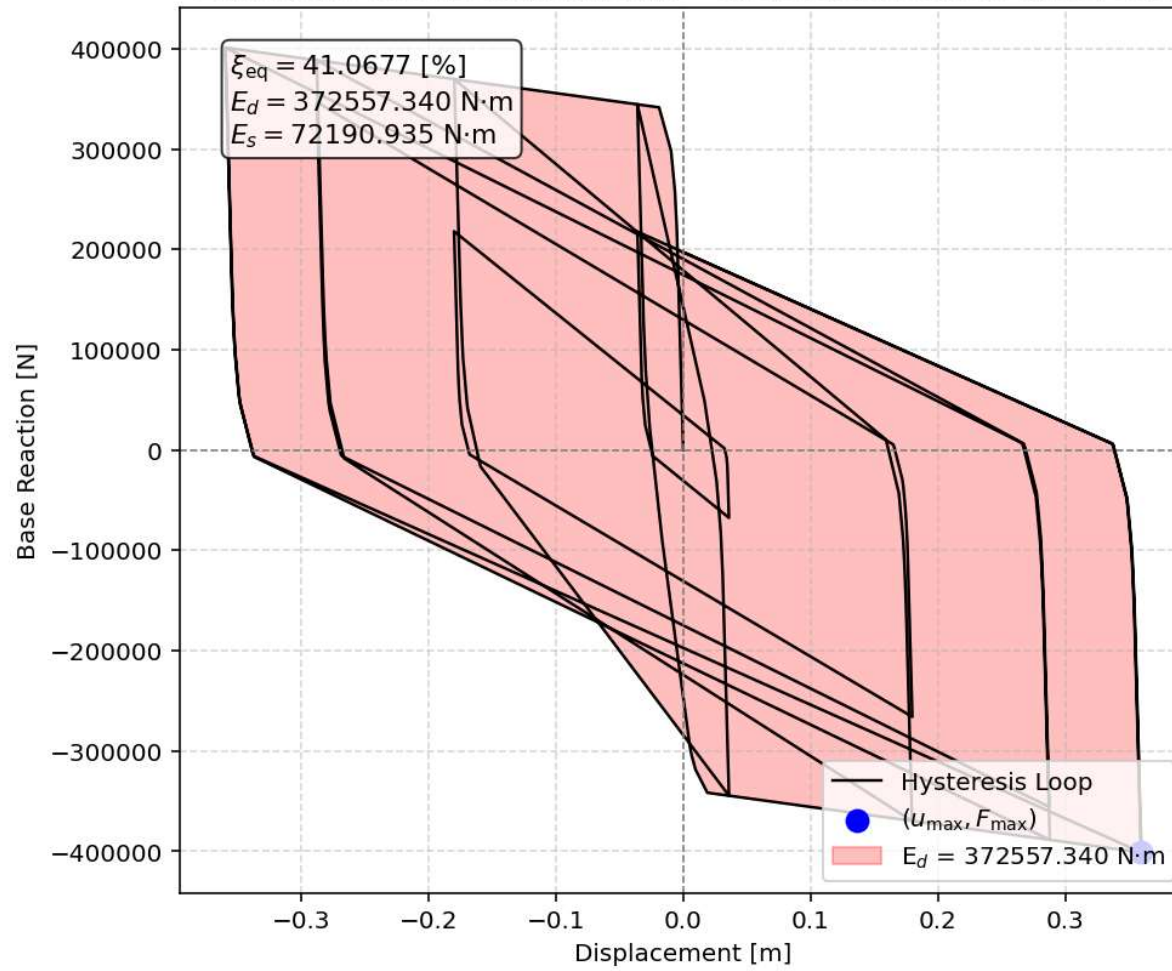




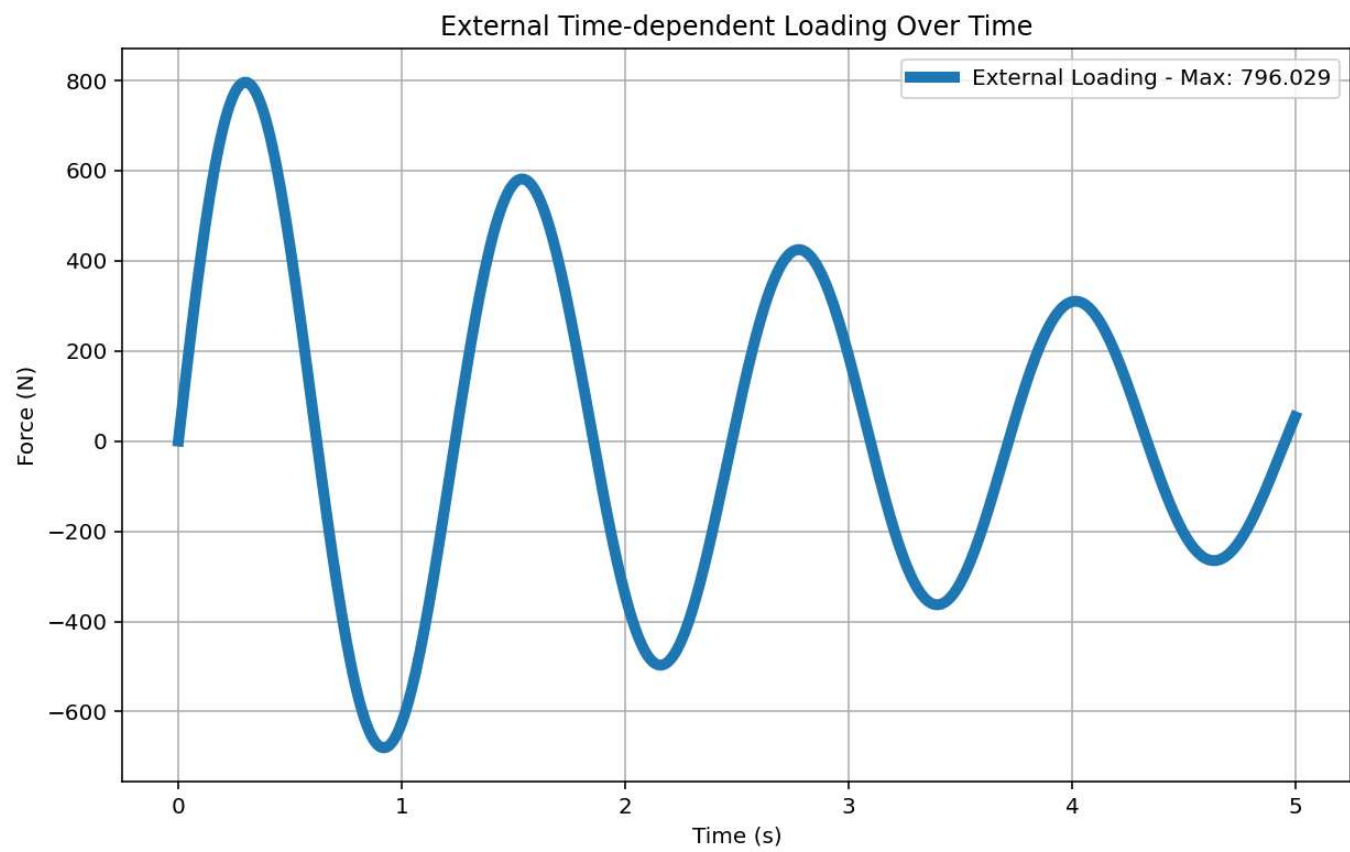


CYCLIC DISPLACEMENT ANALYSIS

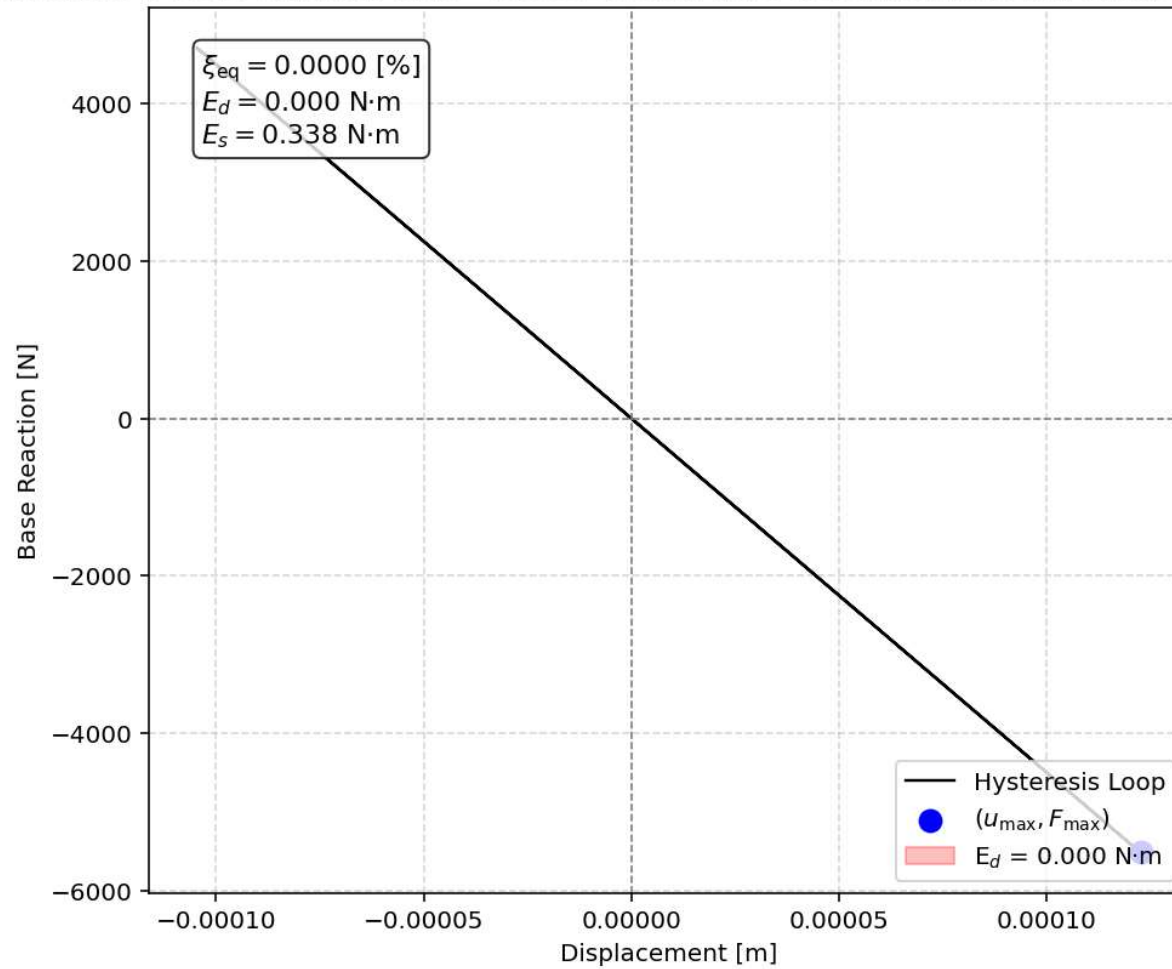
Equivalent viscous damping ratio - Cyclic Displacement Hysteresis



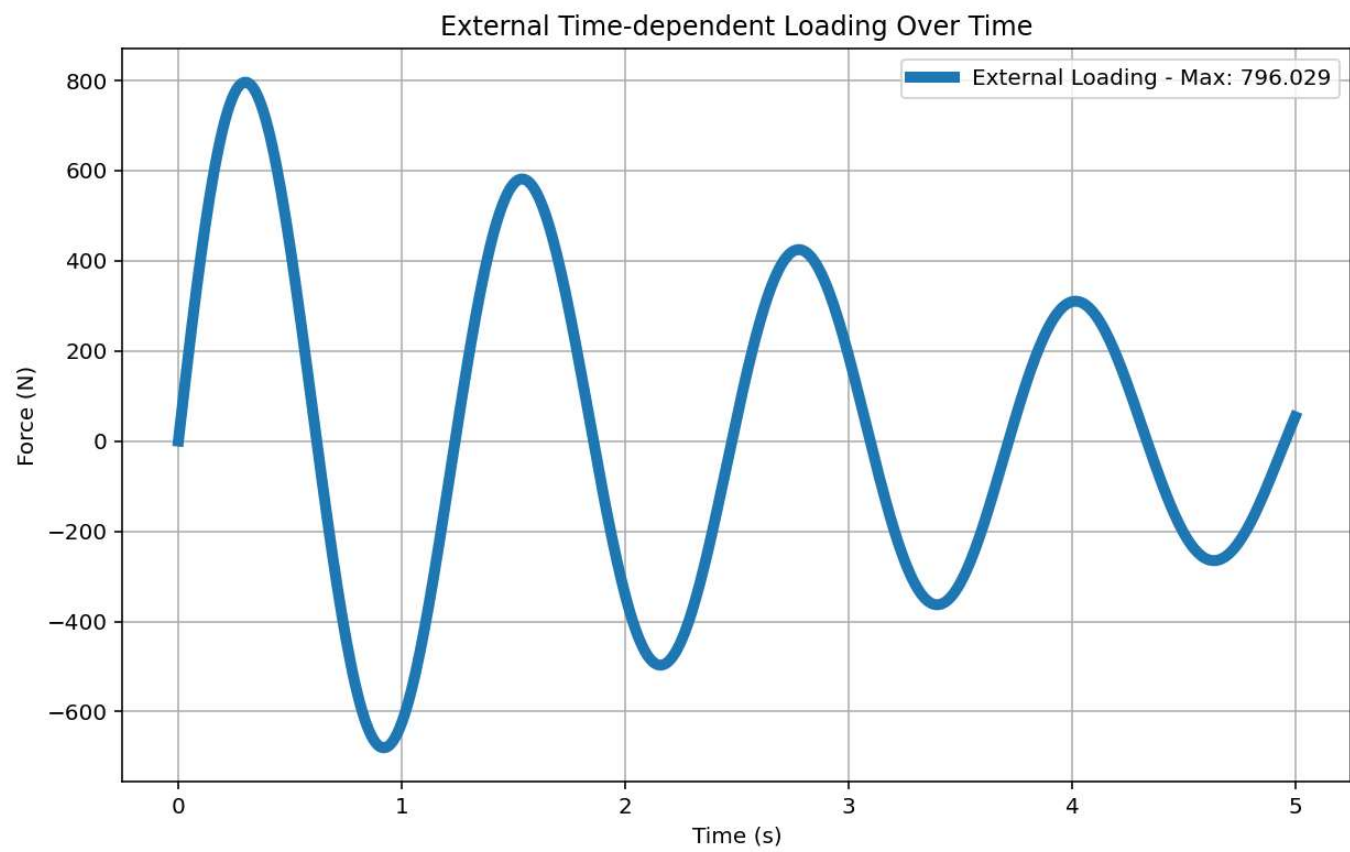
STATIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS



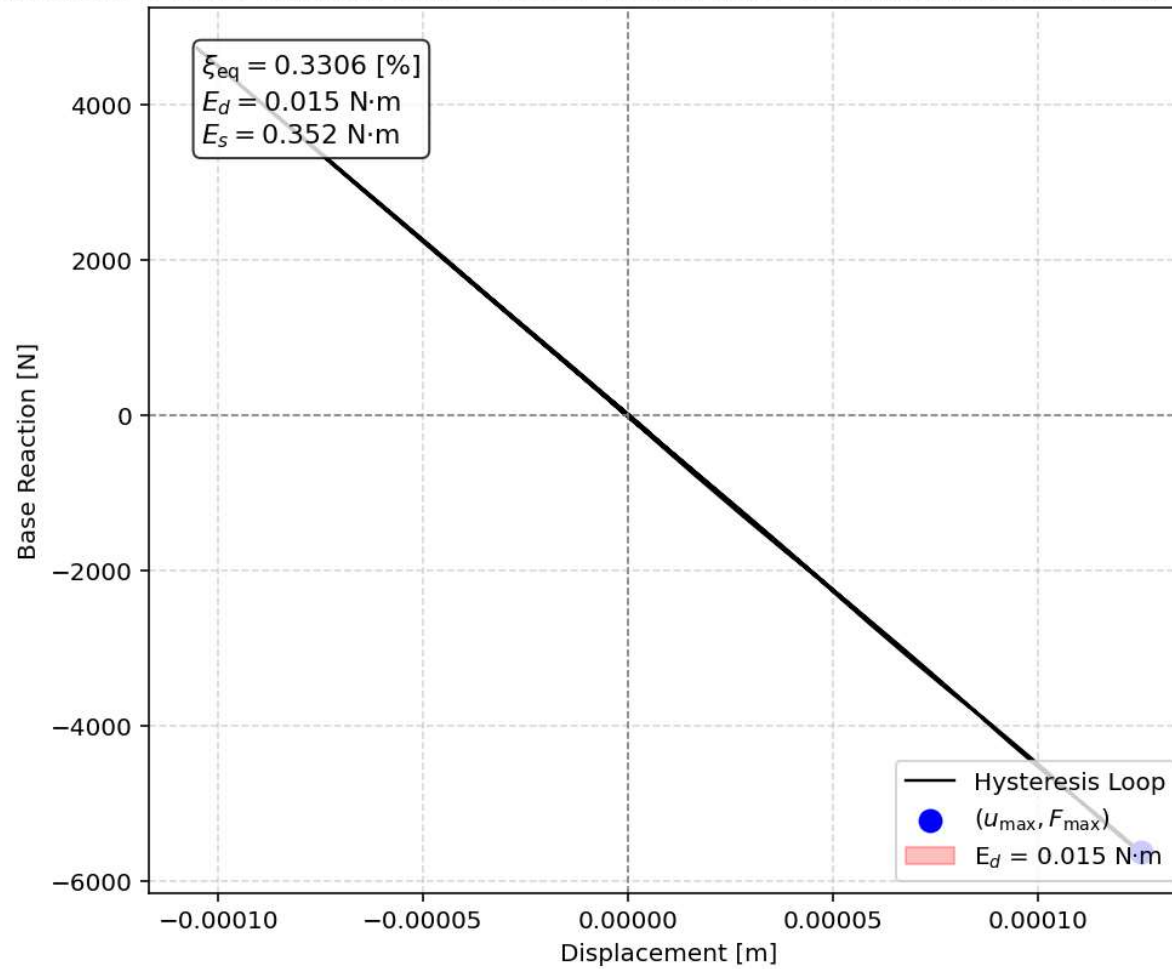
Equivalent viscous damping ratio - Static External Time-dependent Loading Analysis Hysteresis



DYNAMIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

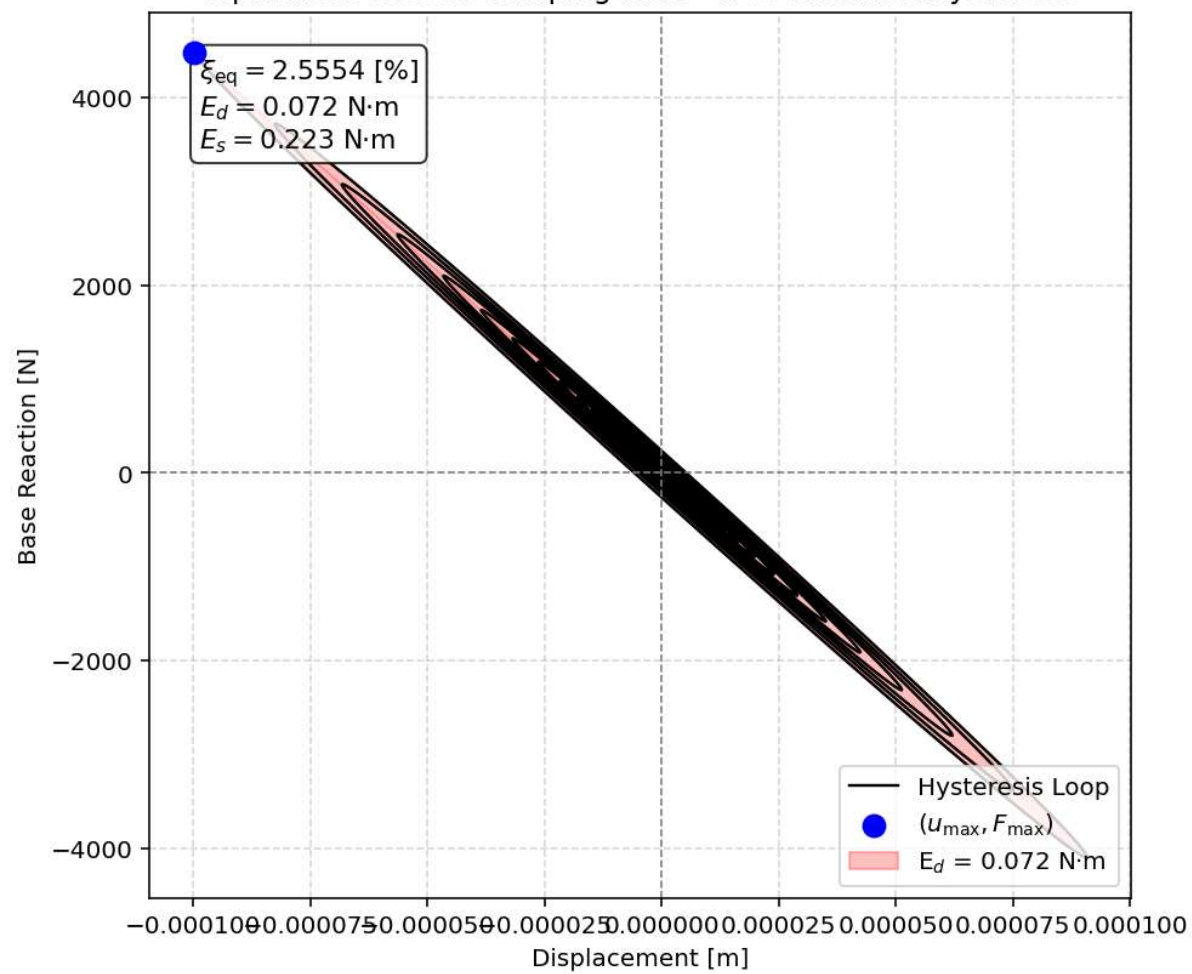


Equivalent viscous damping ratio - Static External Time-dependent Loading Analysis Hysteresis



FREE-VIBRATION ANALYSIS

Equivalent viscous damping ratio - Free-vibration Hysteresis



SEISMIC ANALYSIS

Equivalent viscous damping ratio - Seismic Hysteresis

